

STAFFORD HO SHENG XIAN

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Work Experience

Student Assistant – Research, Singapore Institute of Technology

Sep 2024 – Jan 2025

- Conducted research and achieved breakthroughs in data transmission experiments related to edge intelligence and cloud connectivity using 4G, Bluetooth, and 2.4GHz WiFi.
- Developed an Automated Dispensing System using Python on Raspberry Pi 4 with a user-friendly, scalable interface adaptable for real-time adjustments.

Student Assistant – Projects, Singapore Institute of Technology

Aug 2023 – Sep 2023

- Led a team to fabricate a proof-of-concept for a Net Gun Launcher within 4 weeks.
- Created 3D sketches and CAD drawings using SOLIDWORKS.
- Streamlined project workflows and ensured timely deliverables.

Intern – Technology Development, Systems on Silicon Manufacturing Company

Mar 2019 – Aug 2019

- Delivered presentations at department meetings using agile methodology.
- Developed a digital QR code system using Excel Macros, increasing task efficiency by 30%.

Intern – Chief Controls Office, Barclays

Jun 2018, Sep 2018 – Oct 2018

- Completed professional training focused on optimizing soft skills.
- Developed a digital central data repository for file migration, improving efficiency by 40%.

Education

Bachelor of Engineering with Honours in Robotics Systems, Singapore Institute of Technology

Aug 2022 – Est. Aug 2026

- President, Student Management Committee (AY2022/23)
- Year 2 Representative, Student Management Committee (AY2023/24)
- Year 3 Representative, Student Management Committee (AY2024/25)

Diploma in Electronic Systems, Nanyang Polytechnic

Apr 2017 – Apr 2020

Technical Skills

- **Software:** Visual Components, RoboDK, SOLIDWORKS
- **Programming Languages:** C, C++, Python
- **Model-Based Systems Engineering:** Arcadia – Capella MBSE Tool
- **Embedded Systems:** Arduino IDE, STM32Cube IDE
- **Frameworks:** Robot Operating System (ROS, ROS2)

Causes I Care For

- Mummy Yummy
- Food from the Heart
- Willing Hearts